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Comparison of Well Productivity Between Vertical, Horizontal and Hydraulically Fractured Wells in Gas-Condensate Reservoirs

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Abstract

Gas condensate reservoirs usually exhibit complex flow behaviors due to the build-up of condensate banks around the wells when the bottomhole pressure drops below the dew point. The liquid phase accumulation in the near-wellbore region forms a ring that reduces the gas relative permeability. As a result, gas production decreases and the liquid phase which is a significant part of the value of the field remains in the reservoir.

Various solutions have been implemented in order to remediate such a productivity loss. They include drilling horizontal wells instead of vertical wells[1-3], hydraulically fracturing vertical wells before or after the development of the condensate bank, and acidizing after the condensate bank has formed[4-14].

In this work, we used reservoir simulation to quantify the increase in well productivity from different remediation solutions and assess their effectiveness. The empirical correlations needed to model non-Darcy flow and capillary number effects, which are among the main parameters that control gas condensate well performance, were calibrated with actual well test data.

As should be expected, we found that horizontal wells and hydraulically fractured vertical wells improve well productivity. The degree of productivity enhancement, however, depends on well and reservoir parameters such as horizontal well lengths, permeability anisotropy, fracture length and fracture conductivity. Different simulation models were run above and below the dew point pressure at the same reservoir and flow conditions. The results of our simulations show that horizontal wells enhance productivity significantly below the dew point. Performance improvement with hydraulic fractures, on the other hand, depends on facture length and fracture conductivity.

Introduction

Well productivity estimation in gas condensate reservoirs is still a challenge. This is due to the complex compositional changes and phase behaviors which occur when wells are produced below the dew point. Accurate estimates of well deliverability in these cases require accurate evaluations of both gas and condensate effective permeabilities[15]. This is particularly important within the near-wellbore region where the effective permeability to gas and the value of flowing bottomhole pressure of producing wells are affected by the condensate saturation distribution.

Three radial zones with different liquid saturations appear around a gas-condensate well producing below the dew point[16]. Away from the well, an outer region has the initial liquid saturation; next, closer to the well, there is a rapid increase in liquid saturation and a decrease in the gas mobility. Liquid in that region is immobile. Next to the well, an inner region is formed where liquid saturation is higher than critical and both oil and gas phases are mobile. Finally, in the immediate vicinity of the well, there is a region with lower liquid saturation due to capillary number effects, which represents the ratio of viscous to capillary forces[17]. The existence of the fourth region is important because it counters the reduction in productivity caused by liquid drop-out.

The loss of productivity in gas condensate reservoirs due to condensation near the wellbore is well documented in the literature[18-20]. It is more significant in low-permeability reservoirs which experience larger bottomhole pressure drops, and consequently, fall below the dew point pressure faster. Several techniques have been used to decrease or delay pressure drawdowns in gas condensate wells in order to increase or maintain gas well deliverability. These include drilling horizontal wells and fracturing vertical wells.

Hydraulic fracturing is the primary choice for enhancing production in a majority of gas condensate reservoirs throughout the world[4]. The effectiveness of hydraulic fractures depends mainly on the fracture length and on the dimensionless fracture conductivity[13]. A horizontal well can be considered as a limiting case of an infinite-conductivity fracture, with a fracture height equal to the wellbore diameter[21]. Horizontal wells increase reservoir contact area and decrease pressure drawdown.

Many publications have addressed the design and effectiveness of vertical fractures[4-6,8,10,11] in gas-condensate vertical wells, but only a few have considered the application of horizontal wells[22], and only in heterogeneous[1] or highly permeable[2] reservoirs (where condensate blockage is not significant even in vertical wells). The results of these studies appear very optimistic. .

In the present study, we used 1D, 2D and 3D fine-grid, fully compositional reservoir simulations to accurately model the near-well behavior in gas condensate wells and quantify the increase in productivity obtained with horizontal wells and hydraulic fractures. To obtain realistic results, data from an actual lean gas condensate reservoir were used in the empirical correlations needed to model non-Darcy flow and high capillary number effects, which are among the main parameters that control well deliverability below the dew point.

Simulation results for horizontal and hydraulically fractured vertical wells were compared in terms of condensate saturation profiles and production performance. We found that, in tight gas-condensate reservoirs, horizontal well performance was different above and below the dew point pressure. Above the dew point, improvement in well productivity compared to vertical wells is limited because of the high mobility of the gas. Below the dew point, the increase in well performance is significant at early production times then decreases slightly as time increases. The general behavior for fractured vertical wells is similar to that of horizontal wells, but the optimum well performance depends on the fracture design.

Background and Theory

The relationship between pressure drop and flow rate, which controls well deliverability, is complicated in producing gas condensate reservoirs by the existence in the near–wellbore region of inertial or 'non-Darcy flow' effects and high capillary number effects, also referred to as viscous stripping, velocity stripping or positive coupling[23]. Modelling such complex phenomena requires using compositional simulation and, as most of the pressure drawdown usually occurs within 10 ft of the wellbore, very small grids near to the well[24]. This study used fine grid models in Eclipse-300, a fully compositional simulator.

Before presenting the results of the simulation studies for different well and reservoir geometries, we first review the parameters associated with complex well performance calculations in gas condensate wells. We then examine the main parameters that influence the performance of horizontal wells and propped fractured wells. This knowledge is then applied to the design of simulation runs to compare productivity from different well completions.

Main parameters affecting flow in gas condensate reservoirs

Capillary number effects

Most gas-condensate reservoirs are found at near-critical conditions where the interfacial tension between gas and condensate is low[25]. Experimental studies have shown that as interfacial tensions (IFT) decrease, the relative permeability curves become progressively straighter (miscible) whereas the residual fluid saturations decrease[26,27]. IFT, however, is not the only parameter that controls the relative permeability profile. An increase in relative permeability with velocity has been demonstrated in numerous laboratory core-flood experiments[26,28] and actual field data[17,29,30]. Danesh et al[31] were the first to report laboratory experiments results showing improvements of relative permeability in condensate systems with increases in velocity or decreases in interfacial tension. This positive coupling effect has also been identified in numerous well test data from different gas-condensate fields[17,29,30].

High velocities and low IFT's both increase the ratio of viscous to capillary forces and can be represented by a single parameter, called the capillary number (N_c):

$$N_c = \frac{\nu\mu}{\sigma\phi} \tag{1}$$

where μ is gas viscosity, ν the interstitial velocity, ϕ the porosity and σ the interfacial tension between gas and liquid phases.

A number of empirical correlations are available in commercial simulators[25,26,28,32,33] to represent the capillary number effect at the reservoir scale. In Eclipse-300, N_c is used in a weighting function (f)[34]:

$$k_{rj} = f_j(N_c)k_{rj_{(base)}} + [1 - f_j(N_c)]k_{r_{(misc)}} \tag{2}$$

used to interpolate between immiscible, $k_{r_{(base)}}$, and miscible, $k_{r_{(misc)}}$, relative permeability curves. j denotes the phase. The weighting function f_j is defined as:

$$f_j = (N_{c_{(base)}} / N_c)^{\frac{1}{n_{1j} S_j^{n_{2j}}}} \tag{3}$$

where $N_{c(base)}$ (the base capillary number), n_{1j}, and n_{2j} must be determined experimentally for each phase[26]. f_j can take values between 0 (at very high capillary number) and 1 (at very low capillary number). It is possible to use straight-line relative permeabilities as pseudo-relative permeabilities instead of capillary number base models. This, however, seems adequate only for high permeability reservoirs[34].

Capillary numbers must be used for compositional simulations below the dew point otherwise pressure drops and condensate dropout are overestimated and well productivity is underestimated. This is illustrated in Fig. 1, which shows simulations of a drawdown followed by a build up at different N_c values in a low permeability gas condensate reservoir (k=1.5 mD). The corresponding basic reservoir and well parameters are listed in Table 1.

Non-Darcy flow

Darcy's law does not describe gas flow accurately when the flow rate is high. Forchheimer added a term to the Darcy's flow equation to account for inertial effects:

$$\frac{dp}{dl} = (\frac{\mu}{k})\nu + \beta\rho\nu^2 \tag{4}$$

where ρ is the fluid density and β is Forchheimer's non-Darcy coefficient, which is a characteristic of the rock.

β can be determined from multi-rate pressure test analysis or from theoretical or empirical correlations[36]. A typical empirical correlation for single-phase is:

$$\beta = \frac{a}{k^b \phi^c} \qquad (5)$$

where k and ϕ are the permeability and the porosity, respectively and a, b and c are constants.

A limited number of correlations have also appeared in the literature for two-phase flow[37,38], relating β to the fluid saturation and/or the relative permeability:

$$\beta_j = \frac{a}{\phi^b S_j^{\,c} (kk_r)_j^{\,d}} \qquad (6)$$

where S_j is the phase saturation, k_{rj} is the relative permeability and a, b, c and d are constants. These various correlations yield different values for β[39] and have been mostly developed for gas-water systems. The only correlation which considers the positive coupling effects in gas condensate systems and formulates β_g using steady state gas condensate experimental measurements at high velocities is that of Ref. 38. This correlation, however, has not yet been implemented in commercial simulators.

Fig. 2 shows simulations of a drawdown followed by a build up above the dew point pressure in a dry gas reservoir, both with and without inertial effects. β_g is calculated using a correlation by Geertsma[37] and the reservoir and well parameters are the same as in Table 1. The difference between the two simulated drawdown pressures represents the non-Darcy effects, which are constant under single-phase conditions. In the case of two-phase flow, inertial effects create an additional condensate drop-out, because the inertial resistance in one phase is affected by the presence of the second phase. Fig. 3 shows the same simulations as in Fig. 2, but at a flowing bottomhole pressure below the dew point pressure. There is a dramatic increase in inertial effects below the dew point and the pressure drop is increasing with time even at constant gas flow rate. It is however difficult to separate the pressure drop due to inertial effects from that due to liquid drop-out. This is because both non-Darcy and high capillary number effects are rate-dependent functions acting in opposite directions. In order to demonstrate how different results can be obtained with the same correlation for non-Darcy flow, but different exponent values in the interpolation function Eq. 2, the simulations with n_{2g}=-0.5 shown in Fig. 3 were run again with n_{2g}=0 (Fig. 4). Whereas Fig. 3 showed a ten-fold increase in pressure drop due to non-Darcy effects after a production time of 435 days compared with that in a dry gas system (Fig. 1), the increase in Fig. 4 is only 4.6 fold.

Parameters affecting productivity of horizontal wells

The performance of horizontal wells can be expressed by the ratio of horizontal to vertical productivity indices[21]:

$$\frac{J_h}{J_h} = \frac{\ln\left(r_e/r_w\right)}{\ln\left[\dfrac{a+\sqrt{a^2-\left(L/2\right)^2}}{L/2}\right] + \dfrac{h\sqrt{k_h/k_v}}{L\,\ln\left(h\sqrt{k_h/k_v}\,/2r_w\right)}} \qquad (7)$$

where

$$a = (L/2)[0.5 + \sqrt{0.25 + (2r_{eh}/L)^4}\,]^{0.5}$$

Eq. 7 indicates that the productivity of a horizontal well depends on the producing length L and the permeability ratio k_v/k_h. Under steady-state conditions, horizontal wells are more effective than vertical wells in thin reservoirs or in reservoir with high vertical permeability, if the horizontal producing length L_H is greater than $h(k_h/k_v)^{1/2}$. If L_H is equal to $h(k_h/k_v)^{1/2}$, the horizontal behavior is identical to that of a vertical well, whereas if L_H is less than $h(k_h/k_v)^{1/2}$, the horizontal well exhibits a behavior that resemble that of a well with limited entry and its productivity is less than that of a vertical well.

Parameters affecting productivity of hydraulically fractured wells

Hydraulic fracturing involves pumping a fluid at high pressure into the formation to exceed the rock strength and open a fracture in the rock. A propping agent is often pumped into the fracture to keep it from closing when the pumping pressure is released.

The effectiveness of hydraulic fractures is controlled by the fracture length and the fracture conductivity $k_f w_f$, which is the product of the fracture permeability k_f by the fracture width w_f. Hydraulic fractures are characterized by their dimensionless fracture conductivity, defined as the ratio of the fracture conductivity to the product of the reservoir permeability k and fracture half-length x_f[40]:

$$F_{CD} = \frac{k_f w_f}{k x_f} \qquad (8)$$

Under steady-state conditions, the productivity of a fractured well is often calculated with the formula for a vertical well[21]:

$$J = \frac{0.0070708kh/(\mu B)}{\ln(r_e/r_{we})} \qquad (9)$$

by replacing the fractured well with an equivalent vertical well using the effective wellbore radius concept[41]. In Eq. 9, r_e is the drainage radius and r_{we} is the effective wellbore radius. For a high conductivity fracture, the effective wellbore radius is defined as:

$$r_{we}=x_f/2 \qquad (10)$$

For a low conductivity fracture, the effective wellbore radius depends on F_{CD} and x_f[42] (Fig. 5). For a given formation and a given stimulation treatment, F_{CD} and r_{we}/x_f decrease as the fracture length increases, whereas J increases. Consequently, longer fractures are more productive than shorter fractures but the incremental gain in fracture productivity declines as the fracture gets longer[21].

Problem Definition

The main objective of this study is to compare the productivity of horizontal and hydraulically fractured wells with that of vertical wells in gas condensate reservoirs, using a single well compositional model. As discussed in the previous section, productivity of gas condensate wells is controlled by (1) parameters associated with near-well fluid behavior (PVT, capillary number and inertial effects); (2) the specific reservoir properties (for example, the vertical to horizontal permeability ratio); and (3) the well completion (horizontal well or fracture length, proppant flow conductivity, etc.).

Approach

The parameters required for calculating non-Darcy flow effects and near-critical relative permeabilities must be obtained experimentally, but often are not available. Alternatively, they can be obtained from the analysis of well test data[44] when these exhibit a composite behavior below the dew point pressure[17,29,43,45]. Such a composite behavior yields different derivative stabilizations corresponding to the different mobility zones discussed earlier (Fig. 6).

In the present study, we used the relative permeability functions and the parameters for N_c and β_g obtained in a previous paper[43], where we simulated one DST and two production tests from a well in a lean gas condensate field in the North Sea (well CDW1). The field had a condensate gas ratio between 24-33 STB/MMscf and had been developed mainly with slanted and horizontal wells. There was no liquid drop-out during the DST whereas a condensate bank could be identified in production tests a few months later. The tuned parameters obtained for N_c and β_g in Ref. 43 are considered representative of the true values because not only the drawdown and shut-in pressures could be reproduced (Figs 7-9), but the main build-up derivatives could be matched as well, within the limitations of the compositional simulation model (Fig. 10).

PVT

The PVT data were based on the recombined separator samples from another well (CDW5) in the same field. The laboratory experiments on the samples included hydrocarbon analysis, constant composition expansion (CCE), constant volume depletion (CVD) and separator flash test. All experimental results were matched simultaneously in order to develop a representative equation of state (EOS). The modified Peng-Robinson EOS was used to generate the full range of PVT properties needed for input into the simulator. EOS matching with 13 components (originally 25 components) provided good agreement for the dew point pressure (3085 psia calculated vs. 3079 psia measured) and for the maximum condensate liquid drop-out in the CVD experiment (1.9% measured and calculated). Figs 11 and 12 show liquid drop-out comparisons between EOS predicted and measured CVD and CCE data, respectively.

Model gridding

The simulation models used in this study consisted of a single layer homogenous reservoir of 100 ft thickness with a drainage area of 1917 acres, a 10% porosity, a horizontal permeability of 1.5 mD and a vertical to horizontal permeability ratio of 0.1 (typical of sandstone reservoirs[41]).

We used different grid models for different well geometries, with grid sizes increasing logarithmically away from the well in order to decrease CPU and time demands:

(a) vertical well (Fig. 14): a finely gridded 1D radial model was used, with 60 cells in the radial direction. The well was positioned at the centre, along the vertical axis. The innermost grid size was 0.1ft.

(b) horizontal well (Fig. 15): a 3D rectangular Cartesian grid system was used, with 18225 grid cells (27×45×15). The well, parallel to the x direction, is located at the centre of the model and perforated over its entire length.

(c) hydraulically fractured vertical well (Fig. 16): a 2D rectangular Cartesian grid was used, with 5757 cells (57×101). The well with a propped, rectangular, totally penetrating vertical fracture was positioned at the center. The vertical fracture was symmetrical with respect to the well axis, and was parallel to the x axis. In order to accurately model the pressure and influx behavior in the fracture, the cells representing the fracture had variable lengths in the x direction, from 0.2 ft to 13 ft, with the shorter ones near the wellbore and near the fracture tip. In the y direction, the fracture cells had a constant thickness of 0.1 ft (the minimum cell size that provides sufficient stability). With the dimensions fixed, the fracture conductivity is determined only by the fracture permeability.

Where different horizontal or fracture lengths were needed, the longitudinal grid sizes were modified to keep the reservoir volume constant, but the grid aspect ratios and time steps were not changed to reduce numerical diffusion. All models had the same initial gas in place and did not account for wellbore storage or skin. Frictional losses in the wellbore, gravity and capillary pressures were also neglected. One saturation region was used for vertical and horizontal wells, but two were required for fractured vertical wells, one representing the reservoir where relative permeabilities are modified by N_c and inertial effects, the other representing the fracture with straight line relative permeability curves. The relative permeability curves for each region are shown in Fig. 13. Non-Darcy effects were taken into account in the fracture.

Model verification

The simulation models were verified for accuracy by comparing with analytical solutions for single-phase flow. Fig. 17 shows a build-up test for a 700 ft horizontal well: the early radial, linear and pseudo-radial flow regimes exhibited by the analytical solution are well reproduced by the horizontal well simulation model. Similarly, the fracture simulation model matches the behavior of a fracture with a 230 mD conductivity and a 100 ft half-fracture length, as shown in Fig. 18.

Simulation results

Horizontal well vs. non-stimulated vertical well

A first objective of the simulation runs was to compare the pressure drop and the characteristics of the various zones created when the flowing bottomhole pressure drops below the dew point.

A Cartesian plot of pressure vs. time for a 96 day drawdown followed by a 96 day build-up is shown in Fig. 19 for horizontal wells with different lengths (290, 400, 600, 800 and 1000 ft) and for a vertical well. Compositional simulation runs were made with capillary number and inertial effects at a constant gas flow rate of 5 MMscf/D, starting at an initial reservoir pressure of 3200 psia, 120 psi above the dew point pressure. As can be seen in Fig. 19, the vertical well yields a large drawdown whereas the pressure drop in horizontal wells decreases as the horizontal well length increases. The figure shows that horizontal well lengths greater than 290 ft yields lower drawdown than a vertical well in the same reservoirs. This length is within 10% of $h(k_h/k_v)^{1/2}$.

The condensate saturation profile in the reservoir at the end of the drawdown (t_p= 96.4 days) is shown in Fig. 20. For the vertical well, the condensate saturation is simply plotted versus the radial distance from the wellbore. For horizontal wells, the saturation is plotted versus the distance from the heel in the y direction (i.e., perpendicular to the well), as the saturation is not uniform because of permeability anisotropy. Fig. 20 indicates that, as the horizontal well length increases, the condensate saturation decreases at constant flow rate. But, although a horizontal well length of 290 ft was found to be equivalent to a vertical well in Fig. 19 in terms of pressure drawdown, this is not quite true in terms of condensate distribution. As with vertical wells, four condensate saturation regions develop in the reservoir when the bottomhole pressure drops below the dew point pressure in horizontal wells:

(a) Away from the well, the pressure is above the dew point pressure and only a single phase (gas) is present.
(b) Next and towards the well, oil saturation increases but is still below a critical value. The oil phase is thus immobile and only gas flows. The rate of increase in condensate saturation depends on the length of the horizontal wellbore.
(c) Closer to the well, condensate saturation reaches the critical condensate saturation and both oil and gas are mobile but at different velocities. This, however, only happens at short well lengths. For larger lengths, the pressure drawdown is small and the oil saturation remains less than the critical condensate saturation.
(d) Finally, in the immediate vicinity of the wellbore, capillary number effects reduce both the residual and critical oil saturations. This is clearly established in Fig. 20 for the vertical well but only for long well lengths in horizontal wells. For L_{hw}=290 ft, N_c effects are not apparent, although N_c is greater than in longer horizontal wells and the maximum oil saturation is even higher that in the vertical well. This is due to the higher total skin, which increases dramatically as the length of the wellbore decreases.

To compare well productivities between vertical and horizontal wells and to quantify the effectiveness of horizontal wells in gas condensate reservoirs, runs of 5.5 year durations were made above and below the dew point pressure for different horizontal well lengths. The simulations above the dew point were started at an initial reservoir pressure of 5000 psia whereas the initial pressure for the simulations below the dew point was 3200 psi. All runs were at an initial gas production plateau rate of 10 MMscf/D, maintained until a maximum pressure drop of 1700 psi below the initial reservoir pressure was reached. In the runs above the dew point, this guaranteed that the flowing bottomhole pressure remained above 3300 psia, i.e. 220 psi above the dew point pressure.

As a reference, Fig. 21 compares productivity for the vertical well when production is above and below the dew point pressure: at the end of 5.5 years, the loss in the latter is about 48% less than production above. Figs. 22 and 23 then compare gas rate and cumulative gas production, above the dew point for vertical and horizontal wells. The longer the horizontal wellbore length, the longer the duration of the plateau production rate compared to that of the vertical well (Fig. 22). The cumulative production vs. production time is the same as that for a vertical well if the horizontal well length is 290 ft (Fig. 23), and increases with the horizontal well length. A clearer picture is given in Fig. 24, which shows the relative increase in cumulative gas production for each horizontal well length compared to the cumulative production from the vertical well: the increase reaches 30% for L_H=800 and does not improve with longer horizontal wells. This is due to the high mobility of gas and the low pressure differentials in long horizontal wellbores at constant bottomhole pressure.

Figs. 25 and 26 show gas rate and cumulative gas production below the dew point. In this case, the difference in well productivity between horizontal and vertical wells is more significant. The relative increase in cumulative gas production compared to a vertical well exhibits an initial increase followed by a slight decline, not exceeding 15% of the initial value (Fig. 27). The maximum increase is function of the horizontal well length and reaches 80% for L_H=1000 ft compared to 30% above the dew point.

Vertical fractured well vs. non-stimulated vertical well

A series of simulation runs were performed to compare the pressure drop and condensate saturation profiles in fractured vertical wells and in non-stimulated vertical wells. Fractured well models with fractured half lengths of 50 ft (F_{CD}= 3.07), 100 ft (F_{CD}= 1.53), 200 ft (F_{CD}= 0.76) and 300 ft (F_{CD}= 0.5) were used. The fracture incorporated a propping agent with a conductivity of 230 mD.ft.

All runs were similar to the ones for horizontal wells. Identical non-Darcy flow parameters were used in the reservoir and within the fracture and capillary number effects were included for the matrix cells. Figure 28 shows the bottomhole pressure vs. time for an extended drawdown followed by a build-up, for a vertical well and a fractured well with different fracture half-lengths. The models were run at constant gas flow rate of 5 MMscfd and starting from an initial reservoir pressure of 3200 psia. As can be see in Fig. 28, increasing the length of the fracture reduces the pressure drawdown, but the incremental reduction becomes very small for fracture half-lengths greater than 100 ft.

The condensate saturation profile in the reservoir at the end of the drawdown (t_p= 96.4 days) is shown in Fig. 29. For fractured wells, the distance is from the wellbore in the y direction, perpendicular to the fracture plane. Fig. 29 indicates that as the fracture length increases, the condensate saturation decreases at constant flow rate but, as for the pressure drawdowns, the incremental change becomes negligible for fracture half-lengths greater than 100 ft. Fig. 29 also shows that N_c effects are only slightly visible, and only for longer fracture lengths.

To compare well productivity between non-fractured vertical and hydraulically fractured wells, runs of 5.5 year durations were made above and below the dew point pressure for different fracture lengths. Gas rate and cumulative gas production vs. time above the dew point are compared in Figs. 30 and 31. The duration of the plateau production rate (Fig. 30) and the cumulative gas production (Fig. 31) do not change much as a function of the fracture length. The maximum relative cumulative production increase with respect to the vertical well is 30% for all fracture lengths (Fig. 32).

Figs. 33 and 34 show gas rate and gas production below the dew point. The relative increase in cumulative gas production compared to the vertical well (Fig. 35) is similar to that for horizontal wells (Fig. 27), reaching a maximum of 80% for a fracture half-length of 300 ft, then declining by about 15% of the initial value. The incremental percentage improvement in cumulative production decreases as the fracture length increases and one has to consider the cost of creating a longer fracture vs. the benefit of the resulting increased production. In the example of Fig. 35, the difference between the cumulative gas production for 100 and 300 ft fracture half-lengths is only 10% and the optimum choice would be the fracture half-length of 100 ft. The corresponding dimensionless fracture conductivity for this case is F_{CD} =1.53, which is in good agreement with the value F_{CD} =1.6 suggested by Valko and Economides[4].

Discussion

The hydraulic fractures used in this study are ideal fractures, assumed to be uniform in shape, fully penetrating, with uniformly distributed propping agents and straight-line relative permeability curves. In practice, there is little control on the fracture height, width and azimuth., In addition, the fracture fluid may not be fully recovered during clean up after the fracture treatment and the reaming fluids in the fracture may cause the conductivity within the fracture to be non-homogenous. Furthermore, the assumption of straight-line relative permeability curves, considered adequate for high conductivity fractures, may not apply to proppant pack fractures with low conductivities. All these assumptions may therefore introduce significant uncertainties on the results.

Horizontal well vs. Fractured Vertical Well

The compositional simulation results presented above confirms that horizontal wells and hydraulically fractured wells reduce pressure drop and improve well productivity. Fig. 36 shows the resulting relative increase in cumulative gas production below the dew point as a function of the horizontal well length and fracture half-length. For the reservoir and fluid characteristics used in the present study, a fracture with a half-length x_f=50 ft is equivalent to a horizontal well with a length L_{hw}=600 ft (40% improvement over a non-stimulated vertical well after 5 years), and a fracture with a half-length x_f=250 ft is equivalent to a horizontal well with a length L_{hw}=800 ft (65% improvement). The final choice between a horizontal well and the equivalent vertical well with a fracture can only be made from economics.

Conclusions

The main conclusions from this paper are as follows:

1. Well test data can be used to calibrate the parameters of empirical correlations in well performance models when experimental data are not available.
2. Whereas horizontal wells increase productivity in dry gas systems, their performance is even better in gas-condensate reservoirs below the dew point, where they decrease pressure drawdowns and condensate blockage compared to a vertical well.
3. Hydraulically fracturing vertical wells is equally effective in improving productivity in gas-condensate reservoirs below the dew point.
4. The optimum choice between horizontal wells and hydraulically fractured vertical well, when both are technically feasible, can only be made from economic considerations.

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Nomenclature

H = reservoir thickness
J = productivity index
k = permeability
k_h = horizontal permeability
k_r = relative permeability
k_v = vertical permeability
k_f = fracture permeability
L_h = horizontal well length
p = pressure
s = saturation
t_p = production time
β = non-Darcy Coefficient
μ = viscosity
ρ = density
ν = interstitial velocity
w_f = fracture width
$\O$ = porosity
σ = interfacial tension

Abbreviations

BU = build up, the conditions for shutting a well
CCE = constant composition expansion
CDFi = the modified name of a gas condensate reservoir studied in this research
CGR = condensate-gas ratio
CVD = constant volume depletion
DD = drawdown, the conditions for flowing a well
DST = drill stem test
EOS = equation of state
GOR = Gas-oil ratio
PR = Peng-Robinson
N_c = capillary number
nm(p) = rate normalizes gas pseudo pressure
F_{CD} = dimensionless fracture conductivity

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Reservoir Permeability, k	1.5 mD
Average Porosity	10%
Reservoir Temperature, T	183 F
Wellbore Radius, r	0.25 ft
Base capillary Number	1.00E-06
Top Depth	10000 ft
Rock Compressibility	4.00E-06
Net to Gross Ratio	1
Reservoir Thickness	100 ft

Table 1: Well and reservoir properties

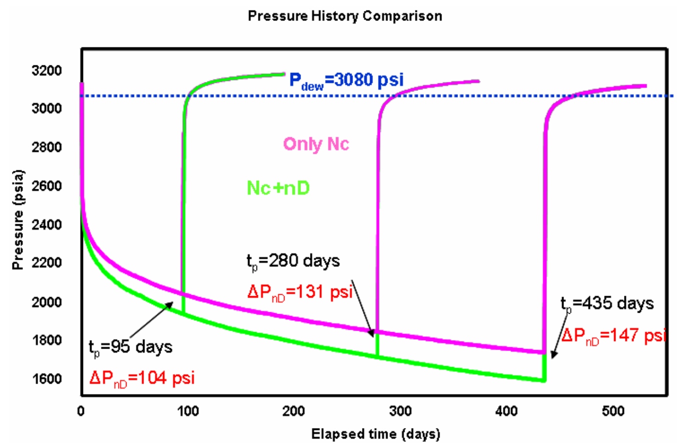

Figure 3: Simulation examples showing increasing pressure drop due to non-Darcy effects below the dew point (n_{2g}=-0.5)

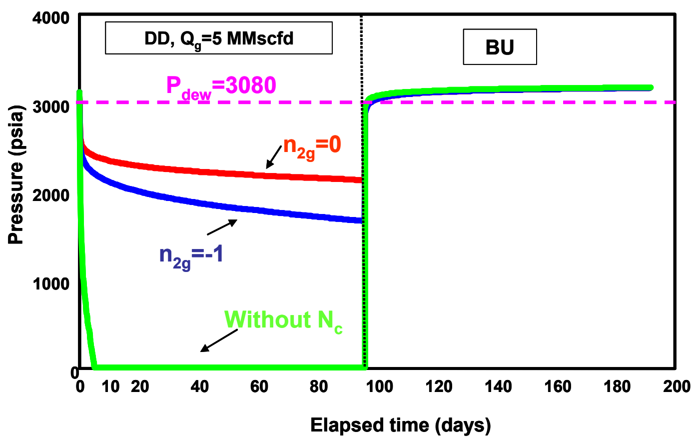

Figure 1: Simulation examples with different N_c

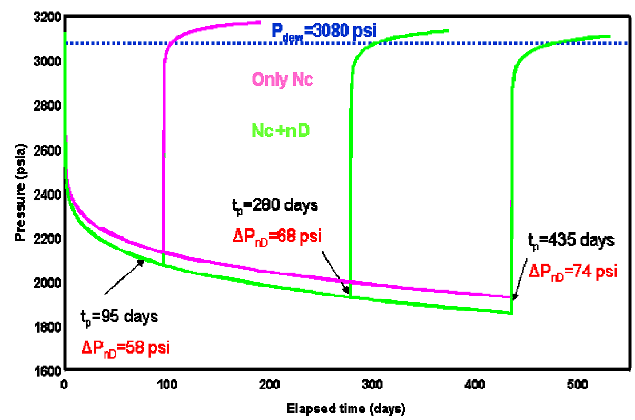

Figure 4: Simulation examples showing increasing pressure drop due to non-Darcy effects below the dew point (n_{2g}=0)

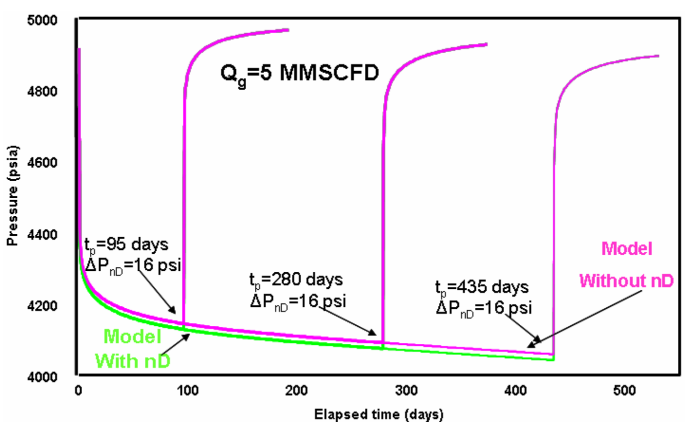

Figure 2: Simulation examples showing the constant pressure drop due to non-Darcy effects above the dew point

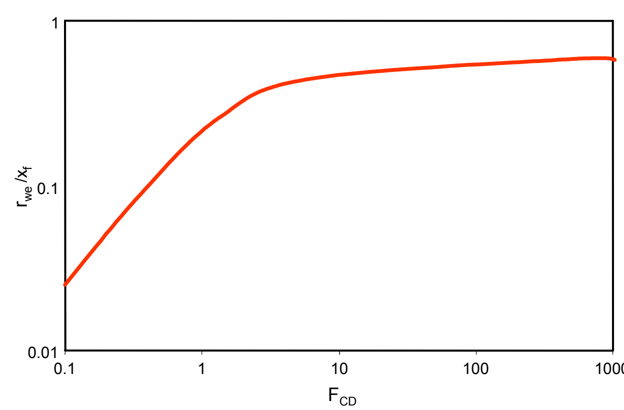

Figure 5: Effective wellbore radius for a finite conductivity fracture[42]

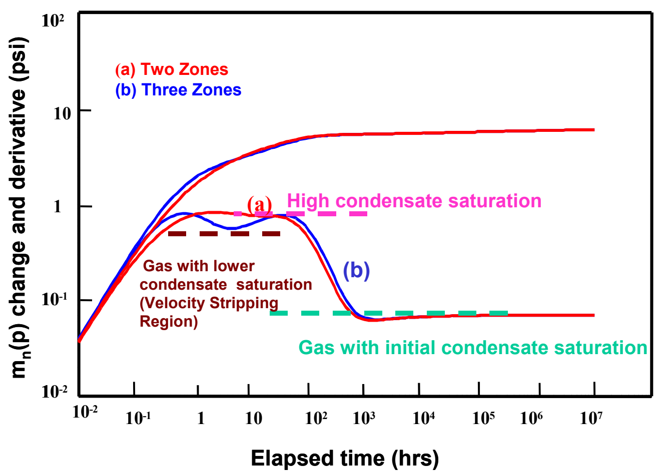

Figure 6: Two- and three-mobility-zone radial composite well test models for vertical wells[17]

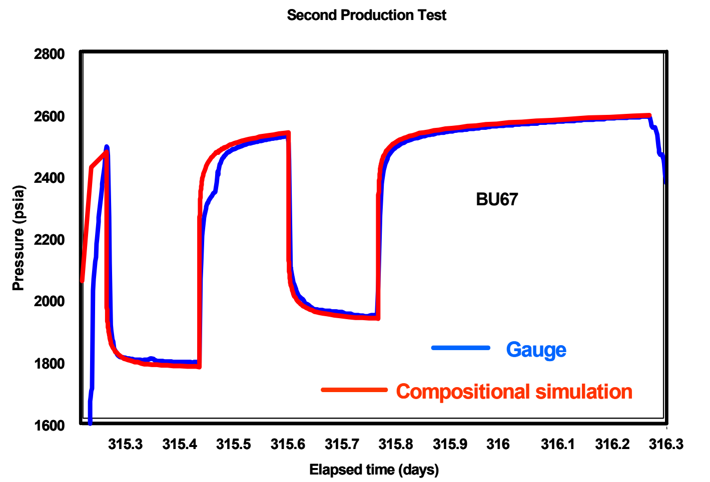

Figure 9: Match between the simulated pressure history and actual pressure data for the second production test in Well CDW1

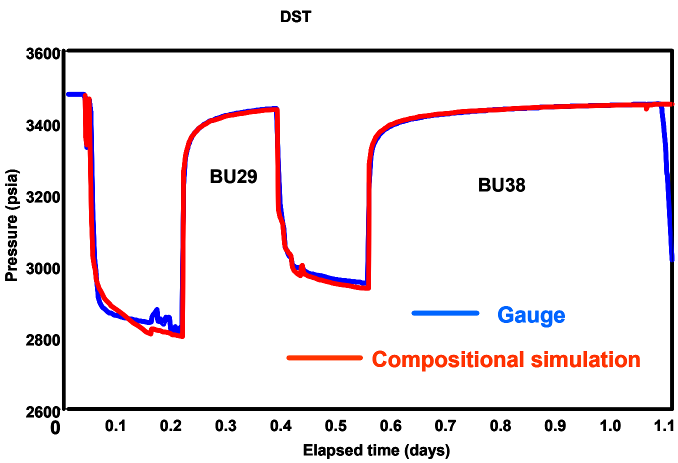

Figure 7: Match between the simulated pressure history and actual pressure data for the DST in Well CDW1

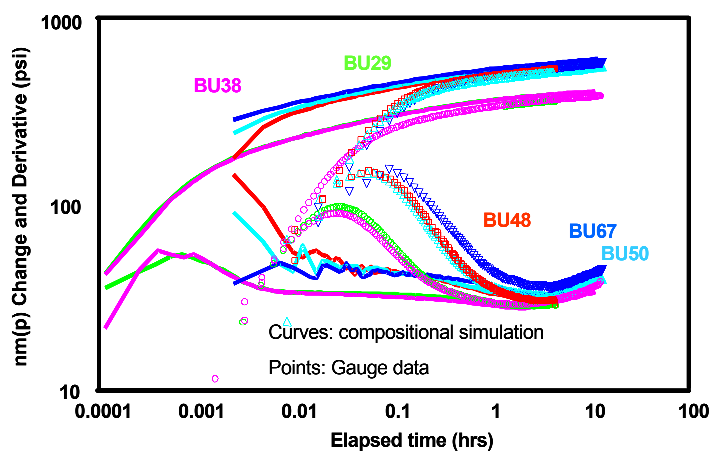

Figure 10: Log-log match of between actual date and compositional simulation responses, Well CDW1

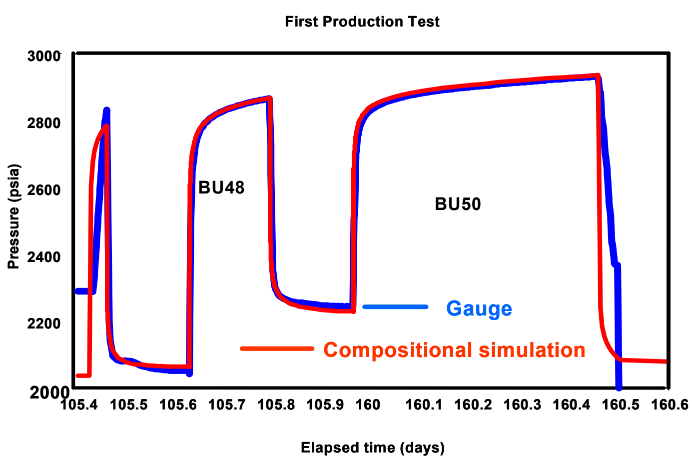

Figure 8: Match between the simulated pressure history and actual pressure data for the first production test in Well CDW1

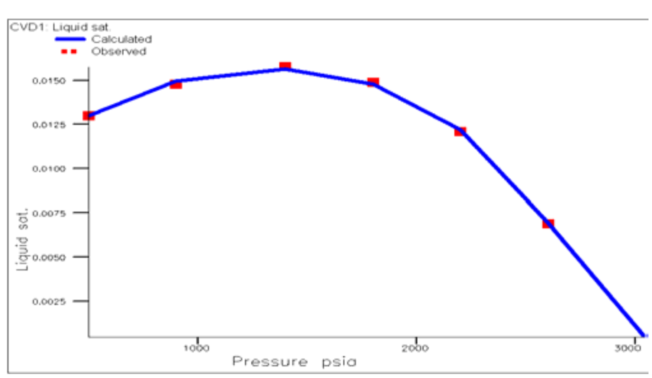

Figure 11: Match between simulated and CVD data, CDFi field

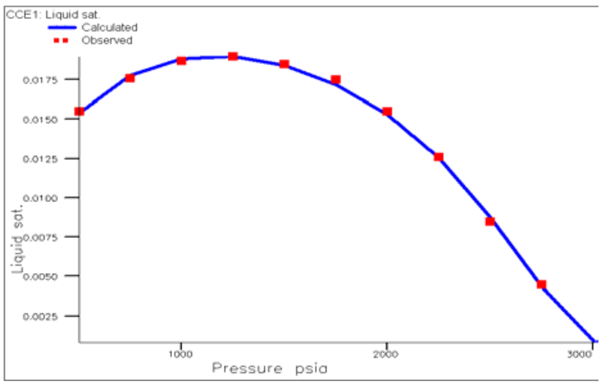

Figure 12: Match between simulated and CCE data, CDFi field

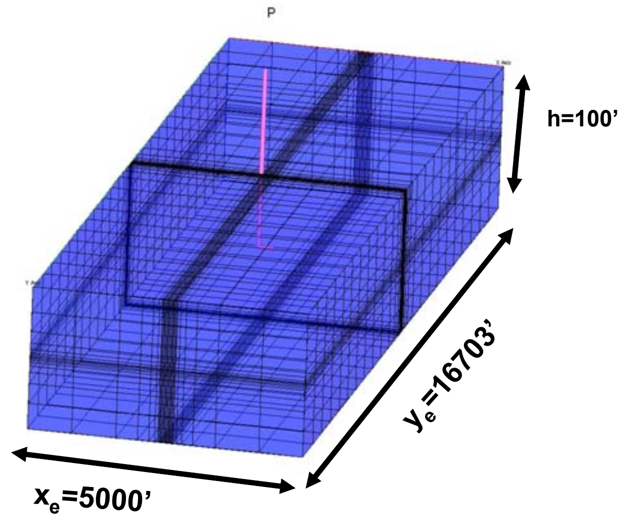

Figure 15: 3D Cartesian grid for horizontal well

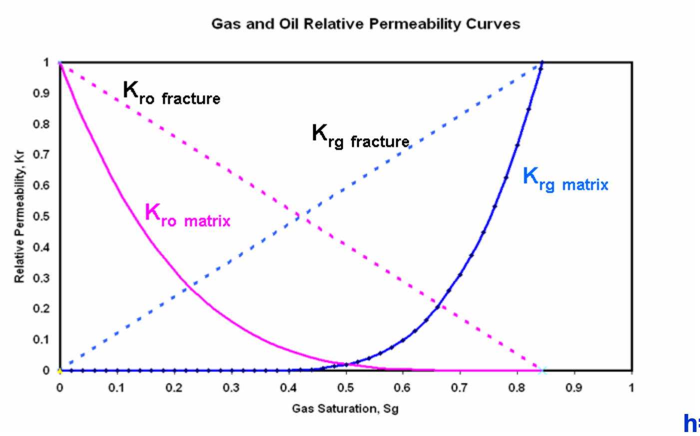

Figure 13: Relative permeability curves

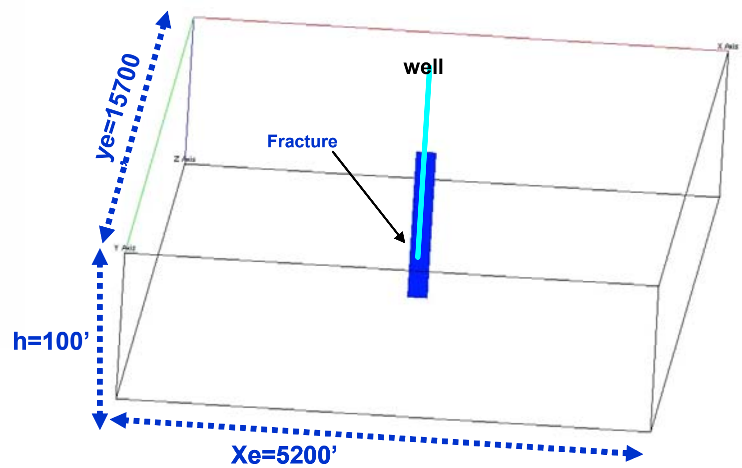

Figure 16: 2D Cartesian grid for fractured vertical well

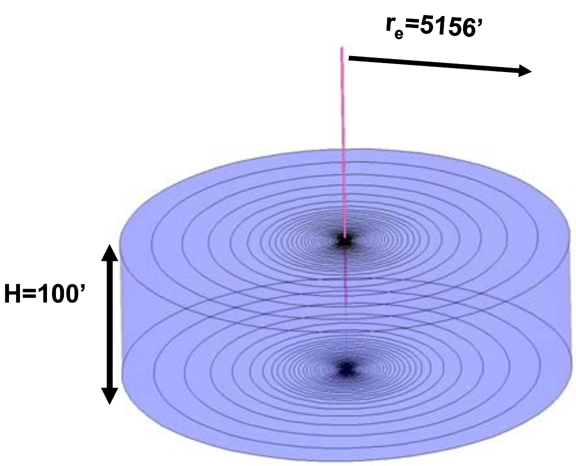

Figure 14: 1D Radial grid for vertical well

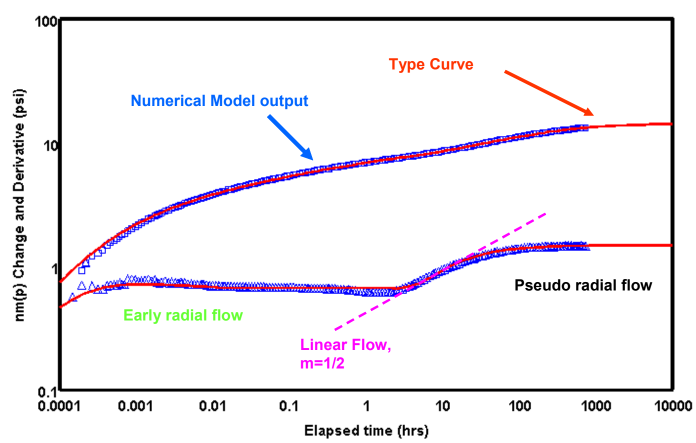

Figure 17: Validation of horizontal well numerical model with analytical solution

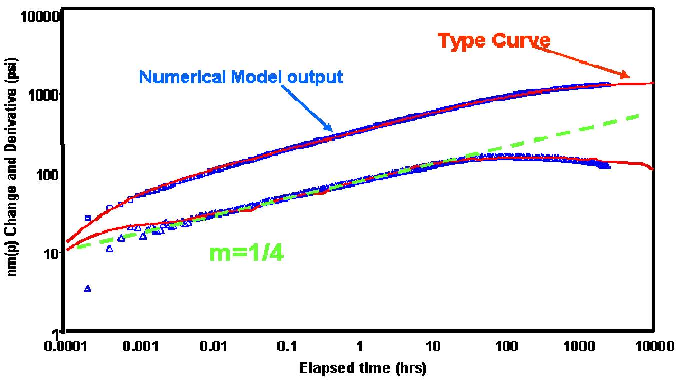

Figure 18: Validation of finite conductivity fractured well numerical model with analytical solution

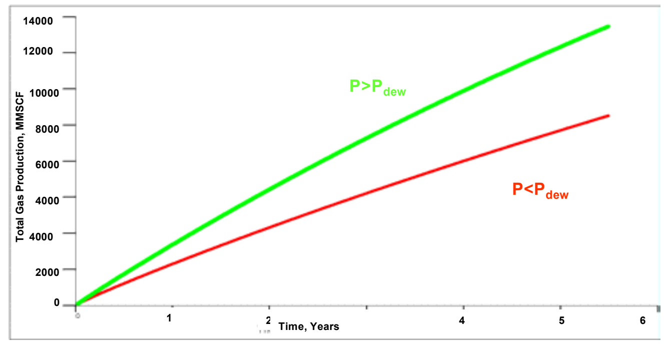

Figure 21: Comparison of cumulative gas productions for a vertical well above and below the dew point

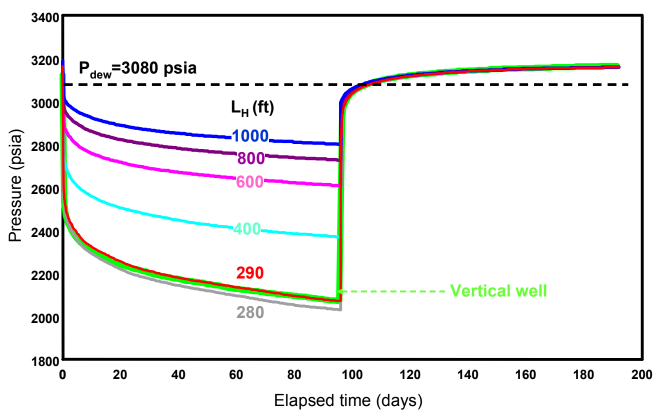

Figure 19: Comparison of pressure drawdowns for vertical and horizontal wells

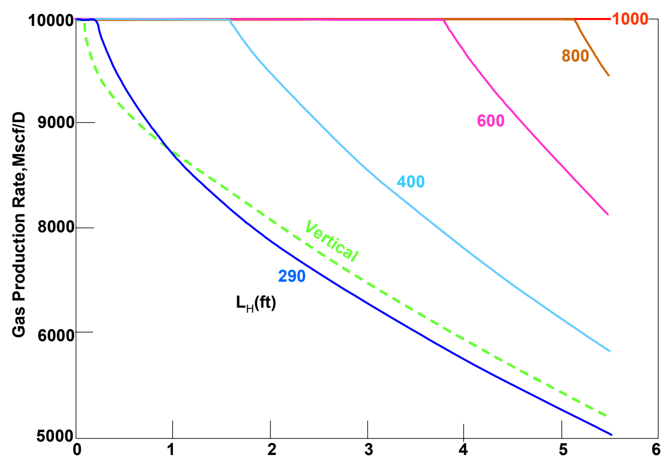

Figure 22: Comparison of gas productions for vertical and horizontal wells above the dew point

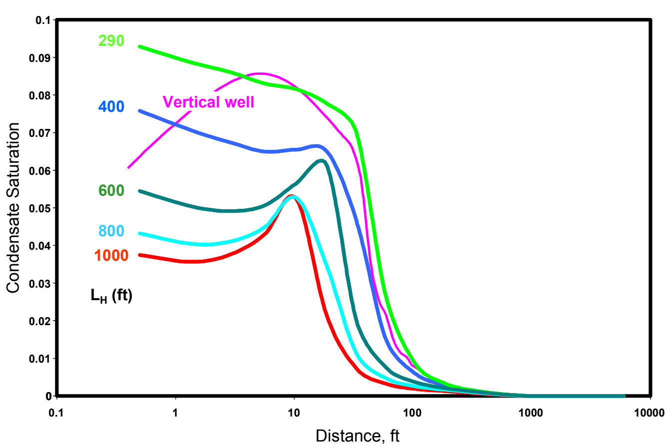

Figure 20: Comparison of condensate saturation profiles for vertical and horizontal wells

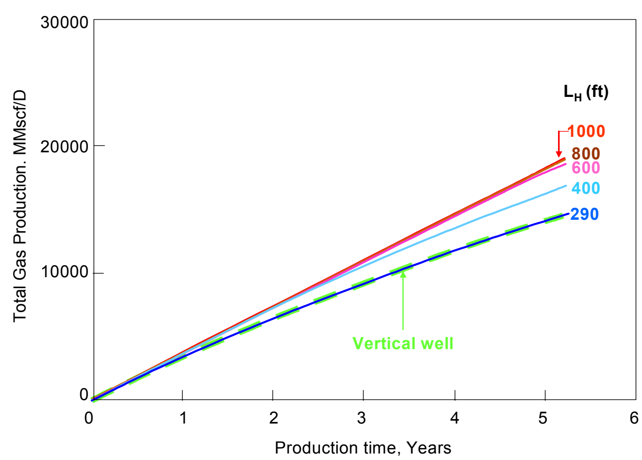

Figure 23: Comparison of cumulative gas productions for vertical and horizontal wells above the dew point

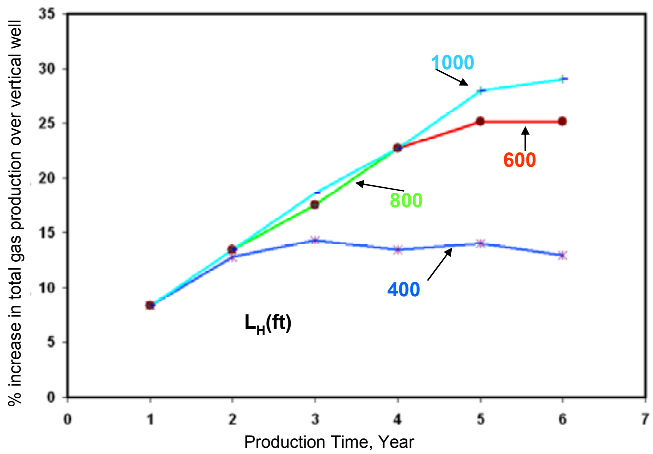

Figure 24: Relative increase in cumulative gas production in horizontal wells compared to that in a vertical well above the dew point

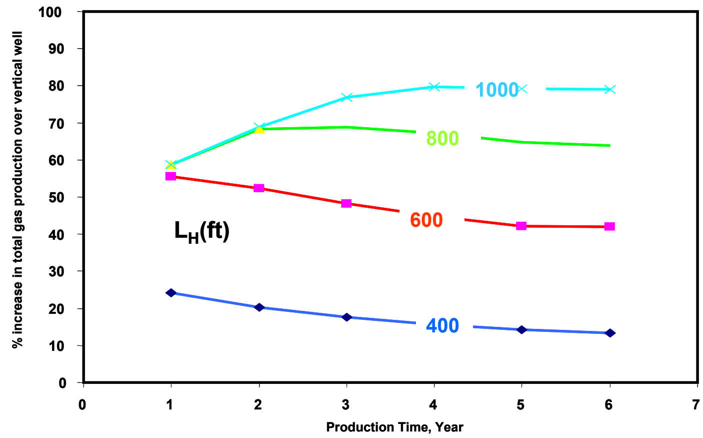

Figure 27: Relative increase in cumulative gas production in horizontal wells compared to that from in a vertical well below the dew point

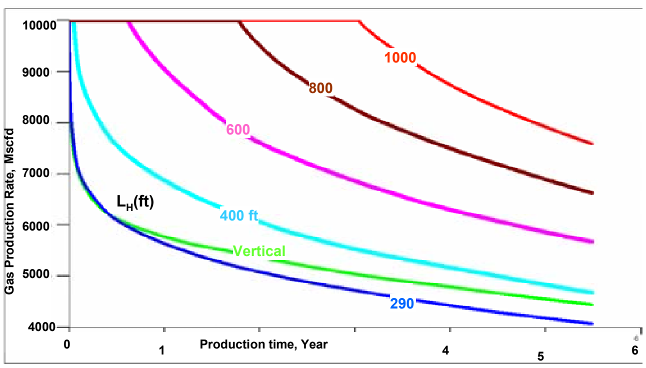

Figure 25: Comparison of gas rates in vertical and horizontal wells below the dew point

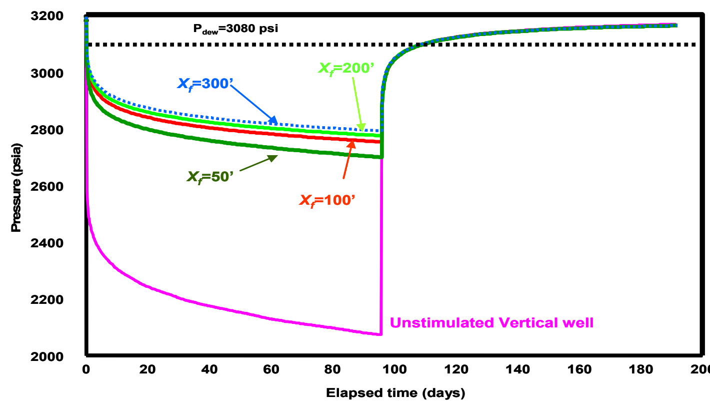

Figure 28: Comparison of pressure drawdowns in vertical, horizontal and fractured vertical wells

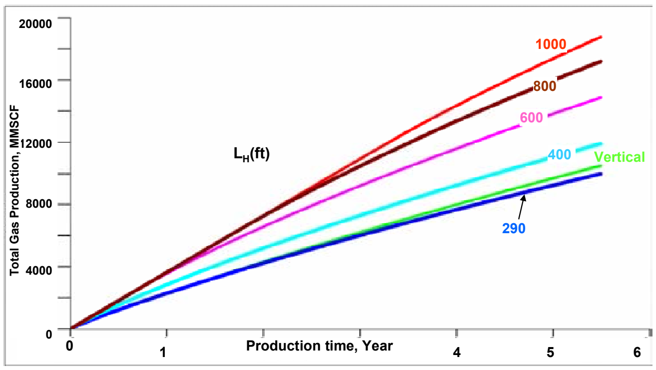

Figure 26: Comparison of cumulative gas production for vertical and horizontal wells below the dew point

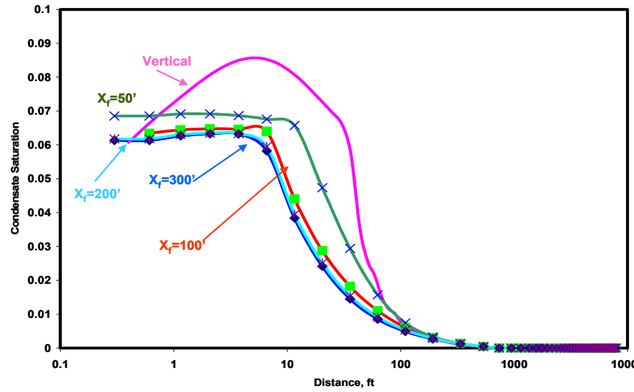

Figure 29: Comparison of condensate saturation profiles in vertical and fractured wells

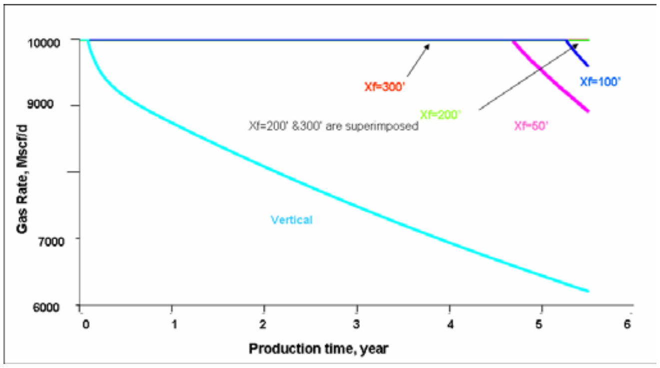

Figure 30: Comparison of gas rates in vertical and fractured vertical wells above the dew point

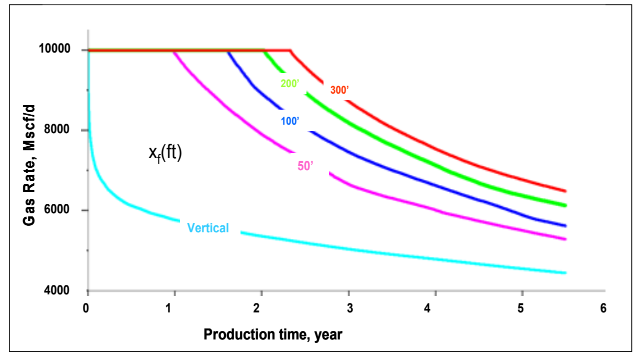

Figure 33: Comparison of gas rates in vertical well and fractured vertical wells below the dew point

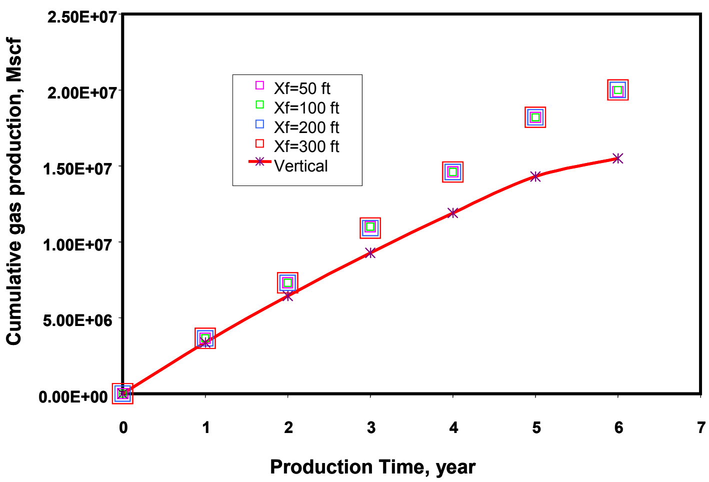

Figure 31: Comparison of cumulative gas productions in vertical and fractured vertical wells above the dew point

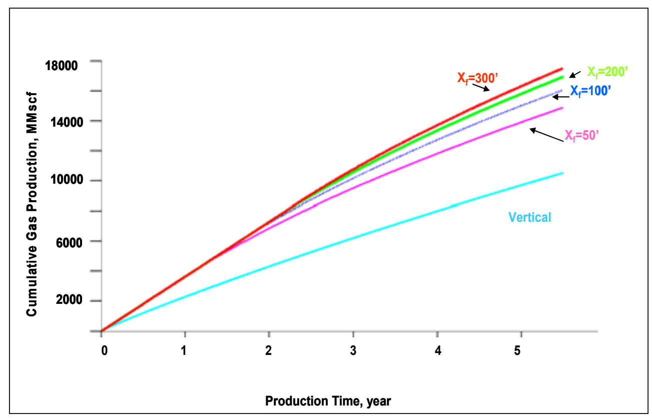

Figure 34: Comparison of cumulative gas productions in vertical and fractured vertical wells below the dew point

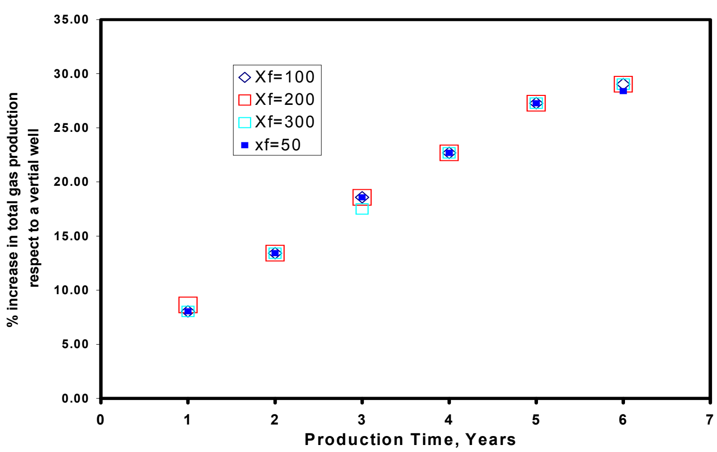

Figure 32: Relative increase in cumulative gas production for fractured vertical wells compared to that for a vertical well above the dew point

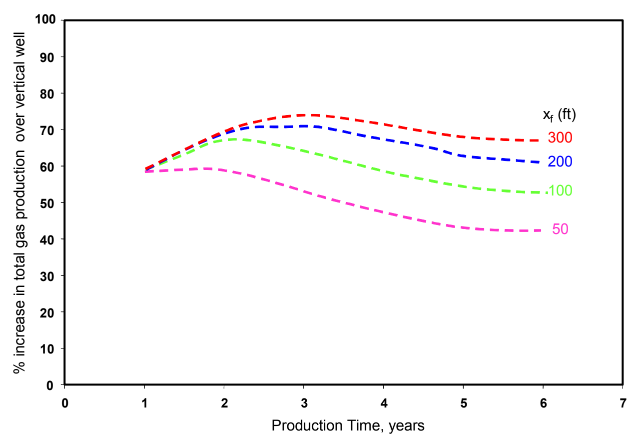

Figure 35: Relative increase in cumulative gas production for fractured vertical wells compared to that from for a vertical well below the dew point

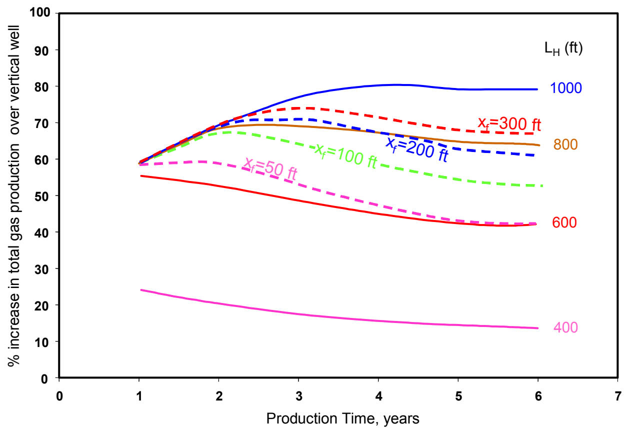

Figure 36: Comparison of relative increase in cumulative gas production for horizontal and fractured vertical wells compared to that from a vertical well below the dew point